

- 2 (a) An object P moves at a constant speed v through an arc of a circle of radius r .

The arc subtends an angle 0.010 rad at the centre of the circle, as shown in Fig. 2.1.

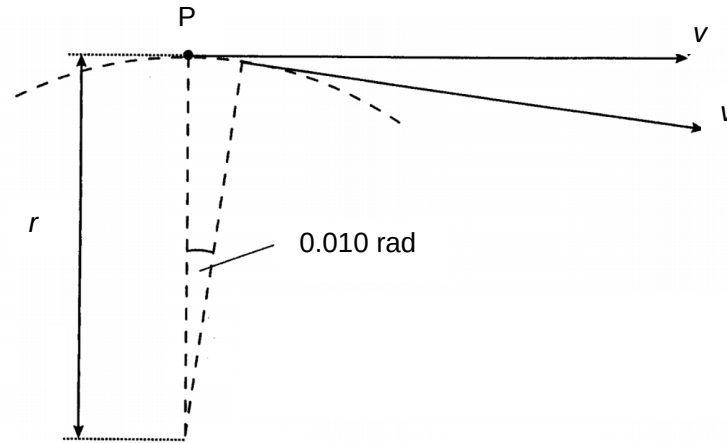


Fig. 2.1

- (i) Sketch a vector diagram to illustrate the change in the velocity of P.

[1]

- (ii) Determine, in terms of v , the magnitude of the change in the velocity of P.

change in velocity = [2]

- (iii) State the direction of the acceleration of P.

..... [1]

(iv) Deduce, in terms of r and v , the time taken for P to travel 0.010 rad.

time taken = [2]

(v) Hence show that the magnitude of the acceleration of P is $\frac{v^2}{r}$.

[1]

(b) Object P is a bob which is spun at a constant high speed around a vertical pole as shown in Fig. 2.2.

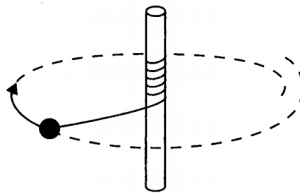


Fig. 2.2

(i) Explain why the thread attached to P may eventually snap.

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 [2]

(ii) Explain whether it is possible for the thread to remain in a horizontal plane during its circular motion.

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 [1]

3 The variation with displacement x of the acceleration a of a 5.0 g particle is shown